

Irrigation & Water Management

AgriMitra AI Knowledge Base — Document 5 of 6 — Water Scheduling by Crop

Irrigation Methods Overview

Method	Best Suited For	Water Efficiency
Flood/Basin irrigation	Paddy, standing-water crops	Low — high water use, simple to manage
Furrow irrigation	Cotton, maize, sugarcane, groundnut	Moderate — better control than flooding
Drip irrigation	Chilli, cotton, sugarcane, vegetables	High — delivers water directly to root zone, saves 30&
Sprinkler irrigation	Groundnut, maize, close-growing crops	Moderate to high — good for light, frequent watering

Signs of Water Stress vs Waterlogging

Condition	Visible Signs
Water stress (deficit)	Leaf rolling/wilting during the day, dull or bluish-green leaf color, delayed flowering, reduced plant he
Waterlogging (excess)	Yellowing of lower leaves, stunted roots, foul smell from soil, algae or moss on the surface, root rot s

Crop-wise Irrigation Guidance

Paddy (Rice)

- Maintain 2–5 cm standing water through most of the vegetative and reproductive stages
- Alternate Wetting and Drying (AWD) technique can reduce water use significantly without major yield loss
- Critical stages needing assured water: active tillering, panicle initiation, and flowering
- Drain the field 7–10 days before harvest

Cotton

- Furrow or drip irrigation preferred over flooding
- Critical stages: squaring, flowering, and boll development — avoid stress at these points
- Reduce irrigation frequency as bolls mature to support opening and picking

Chilli

- Light, frequent irrigation; avoid waterlogging as chilli is highly sensitive to it
- Drip irrigation is well suited and widely recommended
- Critical stages: flowering and fruit development

Maize

- Most sensitive to water stress during tasseling/silking — prioritize irrigation at this stage
- Avoid waterlogging at germination and early seedling stage
- Grain-filling stage also needs consistent moisture for good kernel development

Groundnut

- Critical stages: flowering, pegging, and pod development
- Water stress specifically during pegging directly reduces pod set
- Reduce irrigation near maturity to ease harvesting and reduce pod rot risk

Sugarcane

- Requires frequent irrigation, especially during the grand growth phase (tillering to maturity)
- Drip irrigation increasingly recommended for water-use efficiency in long-duration crop
- Reduce irrigation frequency 3–4 weeks before harvest to improve sugar (brix) content

Note for verification: Exact irrigation intervals depend heavily on local soil type, rainfall, and evapotranspiration rates. Use these as general stage-priority guidance rather than a fixed schedule; local agriculture department advisories provide season-specific schedules.